

PRATEEK GUPTA

Agricultural and Food Engineering (B.Tech + M.Tech Dual 5Y) | 22AG36022

✉ Email | ☎ +91-6378861223 | 🌐 Website | 🔗 LinkedIn | 🐙 GitHub

Education

Degree	Institute	Board/University	CGPA/Percent	Year
B.Tech + M.Tech, AG	IIT Kharagpur	IIT Kharagpur	ongoing	2022–2027
Class XII	Delhi Public School, Bharatpur	CBSE	91.4%	2022
Class X	T.M. International, Bharatpur	CBSE	96.6%	2020

Research Publication

Topological Analysis of Plant Structure Using Persistent Homology for Genotype Classification *Nov 2025*
ACM Conference on Computer Vision, Graphics and Image Processing

- Developed a plant phenotyping framework using **Persistent Homology** and skeleton-based features for structural analysis
- Combined **topological and color features** to improve genotype classification performance using **RF** and **SVM** methods
- Achieved **94%** accuracy using a unified **skeleton + PH setup**, outperforming baseline models by **6%** overall in evaluation

COSINE: Federated Semantic Intrusion Detection with Calibrated Gated Rejection *Jan 2026*
Accepted at IEEE ICSCCC 2026, NIT Jalandhar

- Designed a **semantic, open-set federated IDS** replacing closed-set classification with gated rejection across **5+ classes**
- Embedded network flows and **LLM-derived attack descriptions** into a shared **cosine space** for class-wise reasoning use
- Achieved **97.8%** robust accuracy with **92%+** coverage on the **NSL-KDD** intrusion detection benchmark dataset used

Belief-Aware Spatiotemporal Fault Detection and Recovery in Environmental Sensor Networks *March 2026*
Under Review at Reliability Engineering & System Safety

- Proposed a **belief-aware spatiotemporal framework** for robust fault detection, recovery, and sensor reliability inference
- Developed a **recovery-aware fusion mechanism** to stabilize post-fault transitions and suppress reconstruction oscillations
- Achieved **ROC-AUC of 0.977** with over **70% reconstruction error reduction** under controlled injected fault conditions

Latent Information Instability Framework for Early Crisis Detection *April 2026*
Under Review IEEE Transactions on Computational Social Systems

- Developed a **latent dynamical framework** for robust early crisis detection using multimodal instability signals jointly
- Modeled crisis evolution using **Hidden Markov Models** to capture latent transitions across evolving crisis phases reliably
- Achieved **0.86 early detection rate**, outperforming text-only baselines on real-world disaster benchmark datasets robustly

Internships and Projects

Data Science Intern | Algostrauss *May 2024 – July 2024*

- Achieved an **F1-score of 0.87** in predicting customer churn using RF, XGBoost & LogReg through targeted refinement
- Optimized hyperparameters using **Optuna**, achieving a **5%** increase in predictive accuracy by enhancing model stability
- Engineered **100+** features from customer, transaction, event, and engagement data, achieving a **10%** lift in performance
- Built interactive **Tableau** dashboards to visualize key business metrics for stakeholders, enabling clearer data-driven views

Personalized Credit Card Offer Ranking System | AMEX Hackathon *June 2025 – July 2025*

- Built an offer-ranking system using LightGBM and LambdaRank, raising **MAP@7** from **0.32 to 0.44** through refinements
- Engineered **250+** features from **100M+** customer, transaction, and event rows to support improved ranking optimization
- Applied **SelectKBest** on **600+** raw and engineered features to retain the top **250 predictors**, improving MAP@7 stability
- Tuned **20+** hyperparameters with Optuna for a **15%** lift in recommendation accuracy across multiple customer segments

Technical Skills

Languages	Python SQL Arduino IDE MATLAB Bash/Shell Scripting CSS HTML C C# C++ Rust
ML/DL Libraries	PyTorch TensorFlow Scikit-learn Pandas NumPy Matplotlib Seaborn Prophet Statsmodels
Core Competencies	Machine Learning Deep Learning PCA GNN PINN Time Series Analysis Statistical Analysis RL
Tools & Software	Git GitHub Jupyter Anaconda Tableau FreeCAD Microsoft Office Suite CUDA Label Studio
NLP Architectures	Encoder-Decoder Attention Transformers Transfer-Learning BERT GPT LLMs Llama Mistral
Frameworks	MLflow Docker LangChain Hugging Face OpenCV Autoencoders spaCy PyTorch Lightning

Relevant Coursework & Certifications

- Key Courses:** Programming & Data Structures| DSA, Embedded ML| Probability & Statistics| Intelligent Control
- Certification:** Advanced Machine Learning| Mathematics for Machine Learning Specialization| Bayesian Statistics

Awards and Achievements

- Gold Medal – IIT Kharagpur **General Championship Data Science Competition**, securing 1st place among the institute
- Represented district at the **State Level Lawn Tennis Tournament**, securing 1st Place in District with strong match